$$Y = \frac{X - \mathbb{E}X}{SDX} \qquad \mathbb{E}Y = \mathbb{E}\left(\frac{X - \mathbb{E}X}{SDX}\right) = \frac{1}{SDX}\mathbb{E}(X - \mathbb{E}X) =$$

$$= \frac{1}{SDX} \cdot \underbrace{\left(\mathbb{E}X - \mathbb{E}(\mathbb{E}X)\right)}_{=0} = 0$$

$$SD(Y) = SD\left(\frac{X - \mathbb{E}X}{SDX}\right) = \frac{1}{SDX} \cdot SD(X - \mathbb{E}X) = \frac{1}{SDX} \cdot SDX = 1.$$